A CLOSED SUBSCHEME WHOSE FIBRES ARE QUASI-REGULAR IMMERSIONS IS LOCALLY CONTAINED IN A RELATIVE QUASI-REGULAR IMMERSION, AND AGREES WITH ONE UNDER FLATNESS OR UNDER THE CODIMENSION COUNT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-fibre-quasi-regular is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 5491-5515 and proof span 5517-5594 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof of part (2) is written with the letters of the preceding lemma (it speaks of a closed immersion between the two constructed subschemes and of the corresponding local rings); the wording is the author's and is delivered unchanged, and no letter was normalised by us.

Editorial note. The proof of part (3) contains the author's parenthesis '(some details omitted)' after the statement that the change-of-basis matrix is invertible modulo the maximal ideal; that phrase is kept verbatim.

Editorial note. 10 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The relation between quasi-regularity and regularity of a sequence after localization, and the persistence of a regular sequence in a neighbourhood of a prime. Cited at proof lines 5533-5536. Turns the quasi-regular sequence generating the fibre ideal into a regular sequence on a neighbourhood, which is what the iterative construction needs.
- The fibrewise criterion for a closed subscheme to be a relative effective Cartier divisor, applied one equation at a time, cited from the same chapter. Cited at proof lines 5537-5543. Produces each successive member of the filtration by vanishing loci of the chosen sequence.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- A surjective map of flat local algebras essentially of finite presentation that becomes an isomorphism after dividing by the maximal ideal of the base is an isomorphism, and the local criterion for an isomorphism in a neighbourhood. Cited at proof lines 5554-5559. Upgrades the containment of part (1) to equality in the flat case of part (2).
- The dimension formula at a point of a finite type scheme over a field and the drop of dimension by one on cutting out a nonzerodivisor. Cited at proof lines 5567-5573. Shows that the number of equations cutting out the fibre in the hypothesis of part (3) equals the length of the regular sequence constructed in part (1).

Source reference. Original lemma 00P1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 915–915. Same-chapter reference used in the retained author text at source line(s) 5576. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 00QR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 977–977. Same-chapter reference used in the retained author text at source line(s) 4274, 5560. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 046Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5489–5489. Same-chapter reference used in the retained author text at source line(s) 4271, 5557. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 00KW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 766–766. Same-chapter reference used in the retained author text at source line(s) 3036, 3378, 5578. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 061S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7771–7771. Same-chapter reference used in the retained author text at source line(s) 4873, 5318, 5358, 5533. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 061L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7765–7765. Same-chapter reference used in the retained author text at source line(s) 2970, 4250, 4869, 5359, 5534, 5635. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 062Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7812–7812. Same-chapter reference used in the retained author text at source line(s) 5361, 5538. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's

own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 063U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7843–7843. Same-chapter reference used in the retained author text at source line(s) 5301, 5387, 5483, 5511. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01TR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2384–2384. Same-chapter reference used in the retained author text at source line(s) 4289, 5592. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02FV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3206–3206. Same-chapter reference used in the retained author text at source line(s) 4290, 5593. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $\varphi : X \to S$ be a flat morphism which is locally of finite presentation. Let $T \subset X$ be a closed subscheme. Let $x \in T$ with image $s \in S$.*

1. *If $T_s \subset X_s$ is a quasi-regular immersion in a neighbourhood of x, then there exists an open $U \subset X$ and a relative quasi-regular immersion $Z \subset U$ such that $Z_s = T_s \cap U_s$ and $T \cap U \subset Z$.*
2. *If $T_s \subset X_s$ is a quasi-regular immersion in a neighbourhood of x, the morphism $T \to X$ is of finite presentation, and $T \to S$ is flat at x, then we can choose U and Z as in (1) such that $T \cap U = Z$.*
3. *If $T_s \subset X_s$ is a quasi-regular immersion in a neighbourhood of x, and T is cut out by c equations in a neighbourhood of x, where $c = \dim_x(X_s) - \dim_x(T_s)$, then we can choose U and Z as in (1) such that $T \cap U = Z$.*

In each case $Z \to U$ is a regular immersion by Lemma 063U. In particular, if $T \to S$ is locally of finite presentation and flat and all fibres $T_s \subset X_s$ are quasi-regular immersions, then $T \to X$ is a relative quasi-regular immersion.

Proof. Choose affine open neighbourhoods $\mathrm{Spec}(A)$ of s and $\mathrm{Spec}(B)$ of x such that $\varphi(\mathrm{Spec}(B)) \subset \mathrm{Spec}(A)$. Let $\mathfrak{p} \subset A$ be the prime ideal corresponding to s. Let $\mathfrak{q} \subset B$ be the prime ideal corresponding to x. Let $I \subset B$ be the ideal corresponding to T. By the initial assumption of the lemma we know that $A \to B$ is flat and of finite presentation. The assumption in (1) means that, after shrinking $\mathrm{Spec}(B)$, we may assume $I(B \otimes_A \kappa(\mathfrak{p}))$ is generated by a quasi-regular sequence of elements. After possibly localizing B at some $g \in B$, $g \notin \mathfrak{q}$ we may assume there exist $f_1, \ldots, f_r \in I$ which map to a quasi-regular sequence in $B \otimes_A \kappa(\mathfrak{p})$ which generates $I(B \otimes_A \kappa(\mathfrak{p}))$. By Algebra, Lemmas 061S and 061L we may assume after another localization that $f_1, \ldots, f_r \in I$ form a regular sequence in $B \otimes_A \kappa(\mathfrak{p})$. By Lemma 062Y it follows that $Z_1 = V(f_1) \subset \mathrm{Spec}(B)$ is a relative effective Cartier divisor, again after possibly localizing B. Applying the same lemma again, but now to $Z_2 = V(f_1, f_2) \subset Z_1$ we see that $Z_2 \subset Z_1$ is a relative effective Cartier divisor. And so on until one reaches $Z = Z_n = V(f_1, \ldots, f_n)$. Then $Z \to \mathrm{Spec}(B)$ is a regular immersion and Z is flat over S, in particular $Z \to \mathrm{Spec}(B)$ is a relative quasi-regular immersion over $\mathrm{Spec}(A)$. This proves (1).

To see (2) consider the closed immersion $Z \to D$. The surjective ring map $u : \mathcal{O}_{D,x} \to \mathcal{O}_{Z,x}$ is a map of flat local $\mathcal{O}_{S,s}$-algebras which are essentially of finite presentation, and which becomes an isomorphisms after dividing by $\mathfrak{m}_s$. Hence it is an isomorphism, see Algebra, Lemma 046Y. It follows that $Z \to D$ is an isomorphism in a neighbourhood of x, see Algebra, Lemma 00QR.

To see (3), after possibly shrinking U we may assume that the ideal of Z is generated by a regular sequence $f_1, \ldots, f_r$ (see our construction of Z above) and the ideal of T is generated by $g_1, \ldots, g_c$. We claim that $c = r$. Namely,

$$\begin{aligned} \dim_x(X_s) &= \dim(\mathcal{O}_{X_s,x}) + \text{trdeg}_{\kappa(s)}(\kappa(x)), \\ \dim_x(T_s) &= \dim(\mathcal{O}_{T_s,x}) + \text{trdeg}_{\kappa(s)}(\kappa(x)), \\ \dim(\mathcal{O}_{X_s,x}) &= \dim(\mathcal{O}_{T_s,x}) + r \end{aligned}$$

the first two equalities by Algebra, Lemma 00P1 and the second by r times applying Algebra, Lemma 00KW. As $T \subset Z$ we see that $f_i = \sum b_{ij} g_j$. But the ideals of Z and T cut out the same quasi-regular closed subscheme of X_s in a neighbourhood of x. Hence the matrix (b_{ij}) mod $\mathfrak{m}_x$ is invertible (some details omitted). Hence (b_{ij}) is invertible in an open neighbourhood of x. In other words, $T \cap U = Z$ after shrinking U.

The final statements of the lemma follow immediately from part (2), combined with the fact that $Z \to S$ is locally of finite presentation if and only if $Z \to X$ is of finite presentation, see Morphisms, Lemmas 01TR and 02FV. $\square$